Machine Learning

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Naive Bayes:

- supervised ML used for classification
- non-parametric model
- based on probability - Bayes Theorem

conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \& \quad P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$P(A \cap B) = P(B \cap A)$$

$$P(A|B).P(B) = P(B|A).P(A)$$

$$\boxed{P(A|B) = \frac{P(B|A).P(A)}{P(B)}}$$ Bayes theorem

Multinomial Naive Bayes:

- all features are categorical in multinomial NB

eg. animal type classification - Tiger vs Elephant vs Horse

height:	S	L	L	M	L	L	M	M	M	L	M	M	M	M
weight:	S	L	S	L	M	S	M	M	L	L	L	S	S	S
class:	T	E	H	E	T	T	H	T	E	E	T	T	E	H

height:	S	L	L	M	L	L	M	M	M	L	M	M	M	M
weight:	S	L	S	L	M	S	M	M	L	L	L	S	S	S
class:	T	E	H	E	T	T	H	T	E	E	T	T	E	H

Frequency Table: used to compute marginal and conditional probabilities

for height

	T	E	H	total
S	1	0	0	1
M	3	3	2	8
L	2	2	1	5
total	6	5	3	14

for weight

	T	E	H	total
S	3	1	2	6
M	2	0	1	3
L	1	4	0	5
total	6	5	3	14

$P(C=T \mid h=M)$

event A $\rightarrow$ $C=T$ event B $\rightarrow$ $h=M$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{P(C=T \text{ and } h=M)}{P(h=M)} = \frac{3/14}{8/14} = \frac{3}{8}$$

$$P(C=T \mid h=L) = \frac{2/14}{5/14} = \frac{2}{5}$$

$$P(C=E) = \frac{n(C=E)}{n(\text{total})} = \frac{5}{14}$$

$$P(W=S \mid C=E) = \frac{P(W=S \text{ and } C=E)}{P(C=E)} = \frac{1/14}{5/14} = \frac{1}{5}$$

$P(C=T \mid W=M) = \frac{2}{3}$ $P(W=L) = \frac{5}{14}$

height = m
weight = s $\longrightarrow$ $\boxed{\text{multinomial NB}}$ $\longrightarrow$ T / E / H

$$\left.\begin{array}{l} P(c=T \mid h=m \text{ and } w=s) \\ P(c=E \mid h=m \text{ and } w=s) \\ P(c=H \mid h=m \text{ and } w=s) \end{array}\right\}$$

category with highest conditional probability will be the model prediction

$P(c=T \mid h=m \text{ and } w=s)$

event A – $c=T$
event B – $h=m$
event C – $w=s$

$$P(c=T \mid h=m \text{ and } w=s) = P(A \mid B \cap C)$$

$$P(A \mid B \cap C) = \frac{P(B \cap C \mid A) \cdot P(A)}{P(B \cap C)}$$

$\longrightarrow$ Bayes theorem

$$P(x \mid y) = \frac{P(y \mid x) \cdot P(x)}{P(y)}$$

$$= \frac{P(B \mid A) \cdot P(C \mid A) \cdot P(A)}{P(B \cap C)}$$

due to conditional independence

$$P(B \cap C \mid A) = P(B \mid A) \cdot P(C \mid A)$$

$$P(A \mid B \cap C) = \frac{P(B \mid A) \cdot P(C \mid A) \cdot P(A)}{P(B \cap C)}$$

$$P(c=T \mid h=m \text{ and } w=s) = \frac{P(h=m \mid c=T) \cdot P(w=s \mid c=T) \cdot P(c=T)}{P(h=m \text{ and } w=s)}$$

$$P(C=T \mid h=m \text{ and } w=s) = \frac{P(h=m \mid C=T) \cdot P(w=s \mid C=T) \cdot P(C=T)}{P(h=m \text{ and } w=s)}$$

$$= \frac{3/6 \times 3/6 \times 6/14}{P(h=m \text{ and } w=s)} = \frac{3/28}{x} = \frac{0.1071}{x}$$

$$P(C=E \mid h=m \text{ and } w=s) = \frac{P(h=m \mid C=E) \cdot P(w=s \mid C=E) \cdot P(C=E)}{P(h=m \text{ and } w=s)}$$

$$= \frac{3/5 \times 1/5 \times 5/14}{P(h=m \text{ and } w=s)} = \frac{3/70}{x} = \frac{0.0428}{x}$$

$$P(C=H \mid h=m \text{ and } w=s) = \frac{P(h=m \mid C=H) \cdot P(w=s \mid C=H) \cdot P(C=H)}{P(h=m \text{ and } w=s)}$$

$$= \frac{2/3 \times 2/3 \times 3/14}{P(h=m \text{ and } w=s)} = \frac{4/42}{x} = \frac{0.09523}{x}$$

max of above 3 probabilities

we can ignore the common denominator

max prob here belongs to

$P(C=T \mid h=m \text{ and } w=s)$

hence model prediction will be $=T$

$$\begin{matrix} h=m \\ w=s \end{matrix} \longrightarrow \boxed{\begin{matrix} NB \\ Clf \end{matrix}} \longrightarrow T$$

Few major issues in NB:

1) Numerical Instability :

$$P(C = c_1 \mid x_1 \cap x_2 \cap x_3 \cap x_4 \ldots x_n)$$
$$\propto \underbrace{P(x_1 \mid c_1) \cdot P(x_2 \mid c_1) \cdot P(x_3 \mid c_1) \cdot P(x_4 \mid c_1) \ldots\ldots P(x_n \mid c_1)}_{\text{product of multiple probabilities causes numerical instability issue.}} \cdot P(c_1)$$

to solve it, we apply log

$$P(C = c_1 \mid x_1 \cap x_2 \cap x_3 \cap x_4 \ldots x_n)$$
$$\propto \log[P(x_1 \mid c_1)] + \log[P(x_2 \mid c_1)] \ldots + \log[P(x_n \mid c_1)] + \log[P(c_1)]$$

2) Zero probability issue:

freq. table for height

	T	E	H	total
S	1	0	0	1
M	3	3	2	8
L	2	2	1	5
total	6	5	3	14

$$\left.\begin{aligned} P(h = S \mid c = E) &= 0/5 = 0 \\ P(h = S \mid c = H) &= 0/3 = 0 \end{aligned}\right\} \text{realistically wrong}$$

zero probabilities are due to lack of data
we need to do correction to avoid these zero probabilities
correction/solution : Laplace smoothing

	T	E	H	total
S	1	0	0	1
M	3	3	2	8
L	2	2	1	5
total	6	5	3	14

$\xrightarrow[\text{add } \lambda = 1]{\text{laplace smoothing}}$

	T	E	H	total
S	2	1	1	4
M	4	4	3	11
L	3	3	2	8
total	9	8	6	23

$p(h = s \mid c = E) = 0$

$p(h = s \mid c = E) = \frac{1}{8}$

Text classification using NB:

text	sentiment
1. its a good restaurant	+ve
2. service was bad	-ve
3. food is great	+ve
4. taste was awesome	+ve
5. tables were not clean	-ve

After text preprocessing

Frequency table

vocab	+ve	-ve
good	1	0
restaurant	1	0
serve	1	0
bad	0	1
food	1	0
great	1	0
taste	1	0
awesome	1	0
tables	0	1
clean	0	1

laplace smoothing

Frequency table

vocab	+ve	-ve	total
good	2	1	3
restaurant	2	1	3
serve	2	1	3
bad	1	2	3
food	2	1	3
great	2	1	3
taste	2	1	3
awesome	2	1	3
tables	1	2	3
clean	1	2	3
total	17	13	30

new doc: service is bad

$\downarrow$ preprocessing

serve bad

$$p(+ve \mid text = serve\ bad) = p(+ve \mid serve \cap bad)$$

$$= \frac{p(serve \cap bad \mid +ve) \cdot p(+ve)}{p(serve \cap bad)}$$

$$= \frac{p(serve \mid +ve) \cdot p(bad \mid +ve) \cdot p(+ve)}{p(serve \cap bad)}$$

$$p(-ve \mid text = serve\ bad) = p(-ve \mid serve \cap bad)$$

$$= \frac{p(serve \cap bad \mid -ve) \cdot p(-ve)}{p(serve \cap bad)}$$

$$= \frac{p(serve \mid -ve) \cdot p(bad \mid -ve) \cdot p(-ve)}{p(serve \cap bad)}$$

$$P(+ve \mid text = serve\ bad) = \frac{P(serve \mid +ve) \cdot P(bad \mid +ve) \cdot P(+ve)}{x}$$

$$= \frac{P(serve \cap +ve)}{P(+ve)} \times \frac{P(bad \cap +ve)}{P(+ve)} \times P(+ve) \Big/ x$$

$$= \frac{\frac{\cancel{2}}{17} \times \frac{1}{\cancel{17}} \times \frac{\cancel{17}}{\cancel{30}\,15}}{x} = \frac{\frac{1}{17 \times 15}}{x}$$

$$P(-ve \mid text = serve\ bad) = \frac{P(serve \mid -ve) \cdot P(bad \mid -ve)\ P(-ve)}{x}$$

$$= \frac{P(serve \cap -ve)}{P(-ve)} \times \frac{P(bad \cap -ve)}{P(-ve)} \times P(-ve) \Big/ x$$

$$= \frac{\frac{1}{13} \times \frac{\cancel{2}}{\cancel{18}} \times \frac{\cancel{18}}{\cancel{30}\,15}}{x} = \frac{\frac{1}{13 \times 15}}{x}$$

based on above numbers

$$P(-ve \mid text = serve\ bad) > P(+ve \mid text = serve\ bad)$$

service was bad $\longrightarrow$ $\boxed{\text{NB clf}}$ $\longrightarrow$ $-ve$